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FIGURE 18-8

Probability of detecting solitary nodules on a chest film as a function of log (conspicuity).
(From Revesz, G., Kundel, H. L., and Graber, M. A.²⁹ Used with permission.)

the background. An example of the influence of conspicuity on the detection of simple objects in an image is illustrated in Figure 18-8.

Truth Tables

An image may or may not portray a particular object, and an observer may or may not detect an object that is portrayed. That is, four responses are possible in the interpretation of an image.

1. True-positive (TP): Observer detects an object that is portrayed in the image.
2. False-positive (FP): Observer detects an object that is not portrayed in the image.
3. True-negative (TN): Observer does not detect an object, and no object is portrayed in the image.
4. False-negative (FN): Observer does not detect an object even though an object is portrayed in the image.

These responses are represented in the *truth table* shown in the margin.

In radiology, a film (or examination) is reported as either positive or negative. The film is positive if an abnormality is detected, and it is negative if no abnormality is seen. An independent evaluation (e.g., a surgical biopsy) of the presence or absence of the abnormality in the patient yields the classifications of TP, FP, TN, and FN.

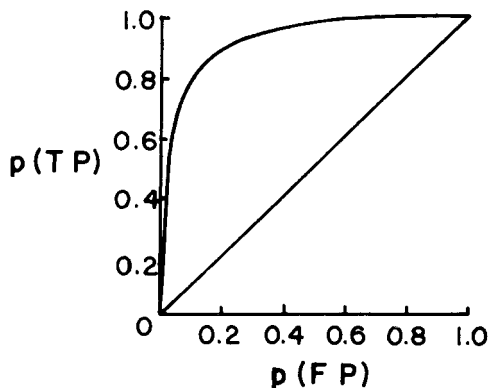
When an object is portrayed in an image, only two responses are possible: The object is either detected or not. That is, $p(\text{TP}) + p(\text{FN}) = 1$, where p represents the probability of each response when the object is present. Similarly, only two responses, FP and TN, are possible when no object is portrayed in the image. Therefore, $p(\text{FP}) + p(\text{TN}) = 1$ when the object is absent. The sensitivity of an imaging procedure is defined as $p(\text{TP})$, the probability of detecting an abnormality when it is present in the patient by examining the images provided by the procedure. The value of $p(\text{TP}) = \text{TP}/(\text{TP} + \text{FN})$, where TP denotes the number of true-positives in a series of responses, and $(\text{TP} + \text{FN})$ signifies the total number of responses when the object of interest is present in the images. The specificity is defined as $p(\text{TN})$, the probability of a negative (normal) interpretation when no abnormality exists in the patient. Often the specificity is defined as $1 - p(\text{FP})$. The value of $p(\text{TN}) = \text{TN}/(\text{TN} + \text{FP})$, where TN denotes the number of true-negatives in a series of responses, and $(\text{TN} + \text{FP})$

Truth Table

Response	Object	
	Present	Absent
Positive	True-positive	False-positive
Negative	False-negative	True-negative

Truth tables require a binary response from the observer, either a positive or negative reaction. This is an example of a two-alternative, forced-choice experiment.

The $p(\text{TP})$ and $p(\text{FP})$ values in an experiment may change depending on how willing or confident the observer is in judging an abnormality.



MARGIN FIGURE 18-4

A representative relative operating characteristics (ROC) curve.

The false-positive probability $p(\text{FP})$ in an ROC curve is frequently expressed as (1-specificity). The true-positive probability $p(\text{TP})$ is then described as the sensitivity.

“The main purpose in reporting an index of performance of a diagnostic system with a sample of cases, a sample of observers, and a sample of readings is not to tell the journal readership how well it performed in this particular sample that was studied, but to provide an estimate of how well it would perform ‘on the average’ in those similar cases and observers and readings that were *not* studied.”³⁰

TABLE 18-1 Expressions Important in Describing the Results of Diagnostic Tests

Expression	Definition ^a
Sensitivity	$\text{TP}/(\text{TP} + \text{FN})$
Specificity	$\text{TN}/(\text{TN} + \text{FP})$
False-positive ratio	$\text{FP}/(\text{FP} + \text{TN})$
Positive predictive value	$\text{TP}/(\text{TP} + \text{FP})$
Negative predictive value	$\text{TN}/(\text{FN} + \text{TN})$
Prevalence	$(\text{TP} + \text{FN})/[\text{TP} + \text{FN} + \text{FP} + \text{TN}]$

^aTP, true-positive; FN, false-negative; TN, true-negative; FP, false-positive.

signifies the total number of responses when the object of concern is not present in the image.

Several other expressions are important to descriptions of the results of diagnostic tests. These expressions are summarized in Table 18-1.

Relative Operating Characteristics Analysis

The performance of observers who are examining images is often described by a relative operating characteristics (ROC) curve. In this approach, the $p(\text{TP})$ (the sensitivity) is plotted against $p(\text{FP})$. An ideal imaging procedure yields a $p(\text{TP})$ of unity and a $p(\text{FP})$ of zero. Data that fall high in the upper left quadrant of an ROC curve depict an accurate imaging procedure. Data that fall on a 45-degree diagonal represent pure guesswork by the observer.

The use of ROC curves is a powerful approach to analysis of the ability of an imaging system to yield definitive information about patients. However, ROC curves are tedious to generate because they must reflect an adequate sampling of all possible images and observers.

SUMMARY

- The three stages of visual perception are
 - Detection
 - Recognition
 - Interpretation
- The principal features of the human eye are the
 - Cornea
 - Iris
 - Lens
 - Retina
- Rod vision provides high sensitivity and low visual acuity.
- Cone vision provides low sensitivity and high visual acuity.
- Models that describe the detection of visual information include:
 - Signal-to-noise model
 - Computational model
 - Signal-channel model
- Visual acuity describes the ability for an observer to extract detailed information from an image. It can be evaluated with a Snellen chart.
- Contrast discrimination describes the ability of an observer to distinguish differences in brightness in an image. It is measured with a contrast discrimination chart.
- Expressions of visual performance include:
 - Contrast detail analysis
 - Conspicuity
 - Truth tables
 - ROC analysis

PROBLEMS

- 18-1. Distinguish between external noise in an image and internal noise in an observer.
- *18-2. An individual receives a visual acuity score of 20/100 with a Snellen chart test. What does the score mean?
- 18-3. Explain the construction of a contrast-detail curve for an imaging system.
- *18-4. Define the expressions “sensitivity” and “specificity” applied to an imaging system.
- 18-5. Describe the construction of an ROC curve.
- *18-6. In screening x-ray mammography examinations, is a high value of sensitivity or specificity considered the more critical value? Why?
- *For problems marked with an asterisk, answers are provided on p. 493.

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ULTRASOUND WAVES

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OBJECTIVES

From studying this chapter, the reader should be able to:

- Explain the properties of ultrasound waves.
- Describe the decibel notation for ultrasound intensity and pressure.
- Delineate the ultrasound properties of velocity, attenuation, and absorption.
- Depict the consequences of an impedance mismatch at the boundary between two regions of tissue.
- Explain ultrasound reflection, refraction and scattering.

INTRODUCTION

Ultrasound is a mechanical disturbance that moves as a pressure wave through a medium. When the medium is a patient, the wavelike disturbance is the basis for use of ultrasound as a diagnostic tool. Appreciation of the characteristics of ultrasound waves and their behavior in various media is essential to understanding the use of diagnostic ultrasound in clinical medicine.¹⁻⁶

HISTORY

In 1880, French physicists Pierre and Jacques Curie discovered the piezoelectric effect.⁷ French physicist Paul Langevin attempted to develop piezoelectric materials as senders and receivers of high-frequency mechanical disturbances (ultrasound waves) through materials.⁸ His specific application was the use of ultrasound to detect submarines during World War I. This technique, sound navigation and ranging (SONAR), finally became practical during World War II. Industrial uses of ultrasound began in 1928 with the suggestion of Soviet Physicist Sokolov that it could be used to detect hidden flaws in materials. Medical uses of ultrasound through the 1930s were confined to therapeutic applications such as cancer treatments and physical therapy for various ailments. Diagnostic applications of ultrasound began in the late 1940s through collaboration between physicians and engineers familiar with SONAR.⁹

WAVE MOTION

A fluid medium is a collection of molecules that are in continuous random motion. The molecules are represented as filled circles in the margin figure (Margin Figure 19-1). When no external force is applied to the medium, the molecules are distributed more or less uniformly (**A**). When a force is applied to the medium (represented by movement of the piston from left to right in **B**), the molecules are concentrated in front of the piston, resulting in an increased pressure at that location. The region of increased pressure is termed a *zone of compression*. Because of the forward motion imparted to the molecules by the piston, the region of increased pressure begins to migrate away from the piston and through the medium. That is, a mechanical disturbance introduced into the medium travels through the medium in a direction away from the source of the disturbance. In clinical applications of ultrasound, the piston is replaced by an ultrasound transducer.

As the zone of compression begins its migration through the medium, the piston may be withdrawn from right to left to create a region of reduced pressure immediately behind the compression zone. Molecules from the surrounding medium move into this region to restore it to normal particle density; and a second region, termed a *zone of rarefaction*, begins to migrate away from the piston (**C**). That is, the compression

In 1794, Spallanzi suggested correctly that bats avoided obstacles during flight by using sound signals beyond the range of the human ear.

“Piezo” is Greek for pressure.

Piezoelectricity refers to the generation of an electrical response to applied pressure.

Pierre Curie used the piezoelectric properties of quartz crystals to construct a device to measure the small changes in mass that accompany radioactive decay. This work was done in collaboration with his wife Marie in her early studies of radioactivity.

The term “transducer” refers to any device that converts energy from one form to another (mechanical to electrical, electrical to heat, etc.). When someone asks to see a “transducer” in a radiology department, they will be shown ultrasound equipment. But strictly speaking, they could just as well be taken to see an x-ray tube.

zone (high pressure) is followed by a zone of rarefaction (low pressure) also moving through the medium.

If the piston is displaced again to the right, a second compression zone is established that follows the zone of rarefaction through the medium. If the piston oscillates continuously, alternate zones of compression and rarefaction are propagated through the medium, as illustrated in **D**. The propagation of these zones establishes a wave disturbance in the medium. This disturbance is termed a *longitudinal wave* because the motion of the molecules in the medium is parallel to the direction of wave propagation. A wave with a frequency between about 20 and 20,000 Hz is a sound wave that is audible to the human ear. An infrasonic wave is a sound wave below 20 Hz; it is not audible to the human ear. An ultrasound (or ultrasonic) wave has a frequency greater than 20,000 Hz and is also inaudible. In clinical diagnosis, ultrasound waves of frequencies between 1 and 20 MHz are used.

As a longitudinal wave moves through a medium, molecules at the edge of the wave slide past one another. Resistance to this shearing effect causes these molecules to move somewhat in a direction away from the moving longitudinal wave. This transverse motion of molecules along the edge of the longitudinal wave establishes shear waves that radiate transversely from the longitudinal wave. In general, shear waves are significant only in a rigid medium such as a solid. In biologic tissues, bone is the only medium in which shear waves are important.

WAVE CHARACTERISTICS

A zone of compression and an adjacent zone of rarefaction constitute one cycle of an ultrasound wave. A wave cycle can be represented as a graph of local pressure (particle density) in the medium versus distance in the direction of the ultrasound wave (Figure 19-1). The distance covered by one cycle is the wavelength of the ultrasound wave. The number of cycles per unit time (cps, or just sec^{-1}) introduced into the medium each second is referred to as the *frequency of the wave*, expressed in units of hertz, kilohertz, or megahertz where 1 Hz equals 1 cps. The maximum height of the wave cycle is the amplitude of the ultrasound wave. The product of the frequency (ν) and the wavelength (λ) is the velocity of the wave; that is, $c = \nu\lambda$.

In most soft tissues, the velocity of ultrasound is about 1540 m/sec. Frequencies of 1 MHz and greater are required to furnish ultrasound wavelengths suitable for diagnostic imaging.

When two waves meet, they are said to “interfere” with each other (see Margin). There are two extremes of interference. In constructive interference the waves are “in phase” (i.e., peak meets peak). In destructive interference the waves are “out of phase” (i.e., peak meets valley). Waves undergoing constructive interference add their amplitudes, whereas waves undergoing destructive interference may completely cancel each other.

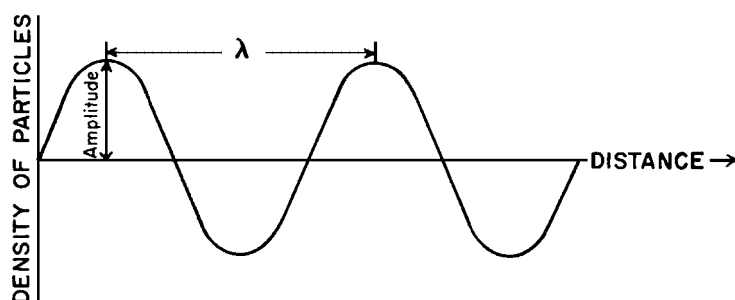
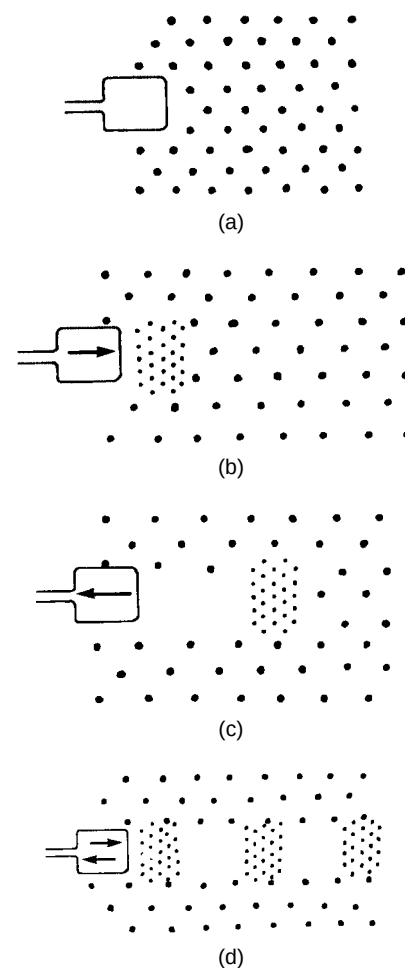


FIGURE 19-1
Characteristics of an ultrasound wave.



MARGIN FIGURE 19-1

Production of an ultrasound wave. **A:** Uniform distribution of molecules in a medium.

B: Movement of the piston to the right produces a zone of compression. **C:** Withdrawal of the piston to the left produces a zone of rarefaction.

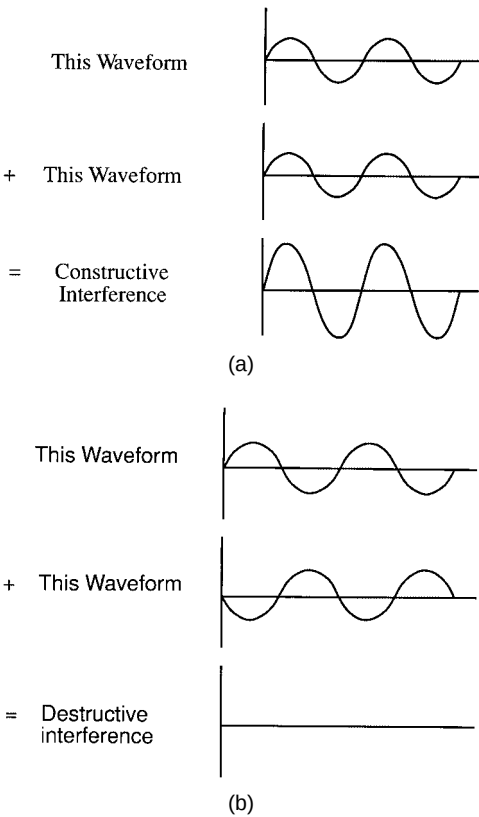
D: Alternate movement of the piston to the right and left establishes a longitudinal wave in the medium.

During the propagation of an ultrasound wave, the molecules of the medium vibrate over very short distances in a direction parallel to the longitudinal wave. It is this vibration, during which momentum is transferred among molecules, that causes the wave to move through the medium.

Frequency Classification of Ultrasound

Frequency (Hz)	Classification
20–20,000	Audible sound
20,000–1,000,000	Ultrasound
1,000,000–30,000,000	Diagnostic medical ultrasound

Ultrasound frequencies of 1 MHz and greater correspond to ultrasound wavelengths less than 1 mm in human soft tissue.



MARGIN FIGURE 19-2
Waves can exhibit interference, which in extreme cases of constructive and destructive interference leads to complete addition (A) or complete cancellation (B) of the two waves.

In the audible range, sound power and intensity are referred to as “loudness.”

Pulsed ultrasound is used for most medical diagnostic applications. Ultrasound pulses vary in intensity and time and are characterized by four variables: spatial peak (SP), spatial average (SA), temporal peak (TP), and temporal average (TA).

Temporal average ultrasound intensities used in medical diagnosis are in the mW/cm² range.

TABLE 19-1 Quantities and Units Pertaining to Ultrasound Intensity

Quantity	Definition	Unit
Energy (<i>E</i>)	Ability to do work	joule
Power (<i>P</i>)	Rate at which energy is transported	watt (joule/sec)
Intensity (<i>I</i>)	Power per unit area (<i>a</i>), where <i>t</i> = time	watt/cm ²
Relationship	$I = \frac{P}{a} = \frac{E}{(t)(a)}$	

■ ULTRASOUND INTENSITY

As an ultrasound wave passes through a medium, it transports energy through the medium. The rate of energy transport is known as “power.” Medical ultrasound is produced in beams that are usually focused into a small area, and the beam is described in terms of the power per unit area, defined as the beam’s “intensity.” The relationships among the quantities and units pertaining to intensity are summarized in Table 19-1.

Intensity is usually described relative to some reference intensity. For example, the intensity of ultrasound waves sent into the body may be compared with that of the ultrasound reflected back to the surface by structures in the body. For many clinical situations the reflected waves at the surface may be as much as a hundredth or so of the intensity of the transmitted waves. Waves reflected from structures at depths of 10 cm or more below the surface may be lowered in intensity by a much larger factor. A logarithmic scale is most appropriate for recording data over a range of many orders of magnitude. In acoustics, the decibel scale is used, with the decibel defined as

$$\text{dB} = 10 \log \frac{I}{I_0} \tag{19-1}$$

where *I*₀ is the reference intensity. Table 19-2 shows examples of decibel values for certain intensity ratios. Several rules can be extracted from this table:

- Positive decibel values result when a wave has a higher intensity than the reference wave; negative values denote a wave with lower intensity.
- Increasing a wave’s intensity by a factor of 10 adds 10 dB to the intensity, and reducing the intensity by a factor of 10 subtracts 10 dB.
- Doubling the intensity adds 3 dB, and halving subtracts 3 dB.

No universal standard reference intensity exists for ultrasound. Thus the statement “ultrasound at 50 dB was used” is nonsensical. However, a statement such as “the returning echo was 50 dB below the transmitted signal” is informative. The transmitted signal then becomes the reference intensity for this particular application. For

TABLE 19-2 Calculation of Decibel Values From Intensity Ratios and Amplitude Ratios

Ratio of Ultrasound Wave Parameters	Intensity Ratio (<i>I</i> / <i>I</i> ₀) (dB)	Amplitude Ratio (<i>A</i> / <i>A</i> ₀) (dB)
1000	30	60
100	20	40
10	10	20
2	3	6
1	0	0
1/2	−3	−6
1/10	−10	−20
1/100	−20	−40
1/1000	−30	−60

audible sound, a statement such as “a jet engine produces sound at 100 dB” is appropriate because there is a generally accepted reference intensity of 10^{-16} W/cm² for audible sound.¹⁰ A 1-kHz tone (musical note C one octave above middle C) at this intensity is barely audible to most listeners. A 1-kHz note at 120 dB (10^{-4} W/cm²) is painfully loud.

Because intensity is power per unit area and power is energy per unit time (Table 19-1), Eq. (19-1) may be used to compare the power or the energy contained within two ultrasound waves. Thus we could also write

$$\text{dB} = 10 \log \frac{\text{Power}}{\text{Power}_0} = 10 \log \frac{E}{E_0}$$

Ultrasound wave intensity is related to maximum pressure (P_m) in the medium by the following expression¹¹:

$$I = \frac{P_m^2}{2\rho c} \quad (19-2)$$

where ρ is the density of the medium in grams per cubic centimeter and c is the speed of sound in the medium. Substituting Eq. (19-2) for I and I_0 in Eq. (19-1) yields

$$\begin{aligned} \text{dB} &= 10 \log \frac{P_m^2/2\rho c}{(P_m^2)_0/2\rho c} = 10 \log \left[\frac{P_m}{P_{m0}} \right]^2 \\ &= 20 \log \frac{P_m}{P_{m0}} \end{aligned} \quad (19-3)$$

When comparing the pressure of two waves, Eq. (19-3) may be used directly. That is, the pressure does not have to be converted to intensity to determine the decibel value. An ultrasound transducer converts pressure amplitudes received from the patient (i.e., the reflected ultrasound wave) into voltages. The amplitude of voltages recorded for ultrasound waves is directly proportional to the variations in pressure in the reflected wave.

The decibel value for the ratio of two waves may be calculated from Eq. (19-1) or from Eq. (19-3), depending upon the information that is available concerning the waves (see Margin Table). The “half-power value” (ratio of 0.5 in power between two waves) is –3 dB, whereas the “half-amplitude value” (ratio of 0.5 in amplitude) is –6 dB (Table 19-2). This difference reflects the greater sensitivity of the decibel scale to amplitude compared with intensity values.

■ ULTRASOUND VELOCITY

The velocity of an ultrasound wave through a medium varies with the physical properties of the medium. In low-density media such as air and other gases, molecules may move over relatively large distances before they influence neighboring molecules. In these media, the velocity of an ultrasound wave is relatively low. In solids, molecules are constrained in their motion, and the velocity of ultrasound is relatively high. Liquids exhibit ultrasound velocities intermediate between those in gases and solids. With the notable exceptions of lung and bone, biologic tissues yield velocities roughly similar to the velocity of ultrasound in liquids. In different media, changes in velocity are reflected in changes in wavelength of the ultrasound waves, with the frequency remaining relatively constant. In ultrasound imaging, variations in the velocity of ultrasound in different media introduce artifacts into the image, with the major artifacts attributable to bone, fat, and, in ophthalmologic applications, the lens of the eye. The velocities of ultrasound in various media are listed in Table 19-3.

The velocity of an ultrasound wave should be distinguished from the velocity of molecules whose displacement into zones of compression and rarefaction constitutes the wave. The molecular velocity describes the velocity of the individual molecules in the medium, whereas the wave velocity describes the velocity of the ultrasound wave

The human ear is unable to distinguish a difference in loudness less than about 1 dB.

Calculation of Decibel Value from Wave Parameters

For X = Intensity in W/cm ² = Power in watts = Energy in joules
use $\text{dB} = 10 \log \frac{X}{X_0}$
For Y = Pressure in pascals or atmospheres = Amplitude in volts
Use $\text{dB} = 20 \log \frac{Y}{Y_0}$

Ultrasound intensities may also be compared in units of nepers per centimeter by using the natural logarithm (ln) rather than the common logarithm (log) where

$$\text{Neper} = \ln (I/I_0)$$

In ultrasound, the term *propagation speed* is preferred over the term *velocity*.

The velocity of ultrasound in a medium is virtually independent of the ultrasound frequency.

The velocity of ultrasound is determined principally by the compressibility of the medium. A medium with high compressibility yields a slow ultrasound velocity, and vice versa. Hence, the velocity is relatively low in gases, intermediate in soft tissues, and greatest in solids such as bone.

TABLE 19-3 Approximate Velocities of Ultrasound in Selected Materials

Nonbiologic Material	Velocity (m/sec)	Biologic Material	Velocity (m/sec)
Acetone	1174	Fat	1475
Air	331	Brain	1560
Aluminum (rolled)	6420	Liver	1570
Brass	4700	Kidney	1560
Ethanol	1207	Spleen	1570
Glass (Pyrex)	5640	Blood	1570
Acrylic plastic	2680	Muscle	1580
Mercury	1450	Lens of eye	1620
Nylon (6-6)	2620	Skull bone	3360
Polyethylene	1950	Soft tissue (mean value)	1540
Water (distilled), 25°C	1498		
Water (distilled), 50°C	1540		

through the medium. Properties of ultrasound such as reflection, transmission, and refraction are characteristic of the wave velocity rather than the molecular velocity.

■ ATTENUATION OF ULTRASOUND

As an ultrasound beam penetrates a medium, energy is removed from the beam by absorption, scattering, and reflection. These processes are summarized in Figure 19-2. As with x rays, the term *attenuation* refers to any mechanism that removes energy from the ultrasound beam. Ultrasound is “absorbed” by the medium if part of the beam’s

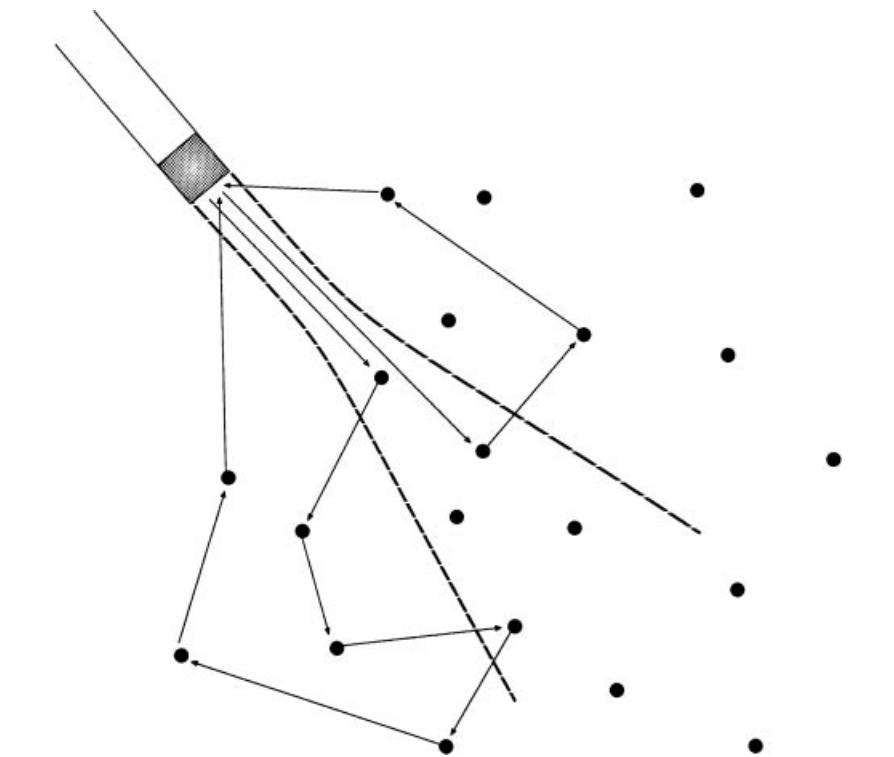
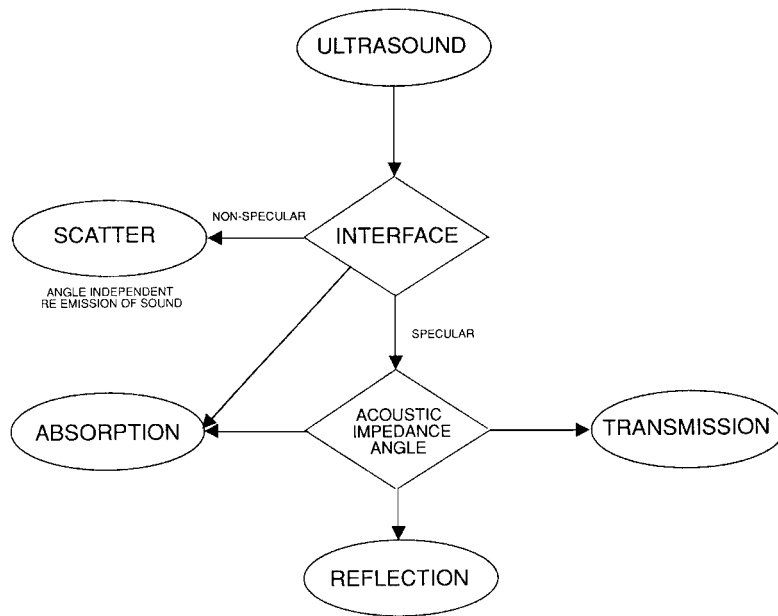


FIGURE 19-2
Constructive and destructive interference effects characterize the echoes from nonspecular reflections. Because the sound is reflected in all directions, there are many opportunities for waves to travel different pathways. The wave fronts that return to the transducer may constructively or destructively interfere at random. The random interference pattern is known as “speckle.”

**FIGURE 19-3**

Summary of interactions of ultrasound at boundaries of materials.

energy is converted into other forms of energy, such as an increase in the random motion of molecules. Ultrasound is “reflected” if there is an orderly deflection of all or part of the beam. If part of an ultrasound beam changes direction in a less orderly fashion, the event is usually described as “scatter.”

The behavior of a sound beam when it encounters an obstacle depends upon the size of the obstacle compared with the wavelength of the sound. If the obstacle’s size is large compared with the wavelength of sound (and if the obstacle is relatively smooth), then the beam retains its integrity as it changes direction. Part of the sound beam may be reflected and the remainder transmitted through the obstacle as a beam of lower intensity.

If the size of the obstacle is comparable to or smaller than the wavelength of the ultrasound, the obstacle will scatter energy in various directions. Some of the ultrasound energy may return to its original source after “nonspecular” scatter, but probably not until many scatter events have occurred.

In ultrasound imaging, specular reflection permits visualization of the boundaries between organs, and nonspecular reflection permits visualization of tissue parenchyma (Figure 19-2). Structures in tissue such as collagen fibers are smaller than the wavelength of ultrasound. Such small structures provide scatter that returns to the transducer through multiple pathways. The sound that returns to the transducer from such nonspecular reflectors is no longer a coherent beam. It is instead the sum of a number of component waves that produces a complex pattern of constructive and destructive interference back at the source. This interference pattern, known as “speckle,” provides the characteristic ultrasonic appearance of complex tissue such as liver.

The behavior of a sound beam as it encounters an obstacle such as an interface between structures in the medium is summarized in Figure 19-3. As illustrated in Figure 19-4, the energy remaining in the beam decreases approximately exponentially with the depth of penetration of the beam into the medium. The reduction in energy (i.e., the decrease in ultrasound intensity) is described in decibels, as noted earlier.

Example 19-1

Find the percent reduction in intensity for a 1-MHz ultrasound beam traversing 10 cm of material having an attenuation of 1 dB/cm. The reduction in intensity (dB) = (1 dB/cm) (10 cm) = −10 dB (the minus sign corresponds to a decrease in intensity

An increase in the random motion of molecules is measurable as an increase in temperature of the medium.

Contributions to attenuation of an ultrasound beam may include:

- Absorption
- Reflection
- Scattering
- Refraction
- Diffraction
- Interference
- Divergence

Reflection in which the ultrasound beam retains its integrity is said to be “specular,” from the Latin for “mirror.” Reflection of visible light from a plane mirror is an optical example of specular reflection (i.e., the original shape of the wave fronts).

An optical example of nonspecular reflection occurs when a mirror is “steamed up.” The water droplets on the mirror are nonspecular reflectors that serve to scatter the light beam.

When scattering objects are much smaller than the ultrasound wavelength, the scattering process is referred to as Rayleigh scattering. Red blood cells are sometimes referred to as Rayleigh scatterers.

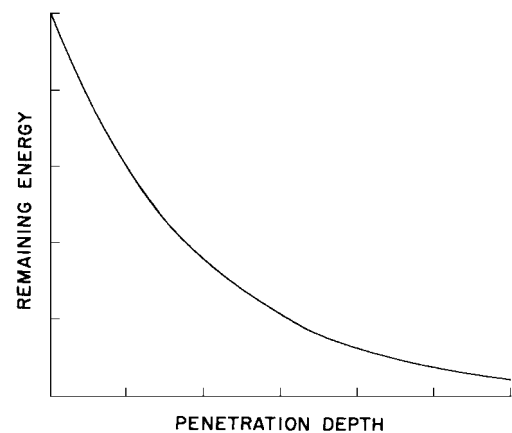


FIGURE 19-4
Energy remaining in an ultrasound beam as a function of the depth of penetration of the beam into a medium.

compared with the reference intensity, which in this case is the intensity of sound before the attenuating material is encountered).

$$\begin{aligned} \text{dB} &= 10 \log \frac{I}{I_0} \\ -10 &= 10 \log \frac{I}{I_0} \\ 1 &= \log \frac{I_0}{I} \\ 10 &= \frac{I_0}{I} \\ I &= \frac{I_0}{10} \end{aligned}$$

There has been a 90% reduction in intensity. Determine the intensity reduction if the ultrasound frequency were increased to 2MHz.

Because the attenuation increases approximately linearly with frequency, the attenuation coefficient at 2 MHz would be 2 dB/cm, resulting in a -20 dB (99%) intensity reduction.

The attenuation of ultrasound in a material is described by the attenuation coefficient α in units of decibels per centimeter (Table 19-4). Many of the values in Table 19-4 are known only approximately and vary significantly with both the origin and condition of the biologic samples. The attenuation coefficient α is the sum of the individual coefficients for scatter and absorption. In soft tissue, the absorption

TABLE 19-4 Attenuation Coefficients α for 1-MHz Ultrasound

Material	α (dB/cm)	Material	α (dB/cm)
Blood	0.18	Lung	40
Fat	0.6	Liver	0.9
Muscle (across fibers)	3.3	Brain	0.85
Muscle (along fibers)	1.2	Kidney	1.0
Aqueous and vitreous humor of eye	0.1	Spinal cord	1.0
Lens of eye	2.0	Water	0.0022
Skull bone	20	Caster oil	0.95
		Lucite	2.0

coefficient accounts for 60% to 90% of the attenuation, and scatter accounts for the remainder.¹¹

Table 19-4 shows that the attenuation of ultrasound is very high in bone. This property, along with the large reflection coefficient of a tissue-bone interface, makes it difficult to visualize structures lying behind bone. Little attenuation occurs in water, and this medium is a very good transmitter of ultrasound energy. To a first approximation, the attenuation coefficient of most soft tissues can be approximated as 0.9ν , where ν is the frequency of the ultrasound in MHz. This expression states that the attenuation of ultrasound energy increases with frequency in biologic tissues. That is, higher-frequency ultrasound is attenuated more readily and is less penetrating than ultrasound of lower frequency.

The energy loss in a medium composed of layers of different materials is the sum of the energy loss in each layer.

Example 19-2

Suppose that a block of tissue consists of 2 cm fat, 3 cm muscle (ultrasound propagated parallel to the fibers), and 4 cm liver. The total energy loss is

$$\begin{aligned}\text{Total energy loss} &= (\text{Energy loss in fat}) + (\text{Energy loss in muscle}) \\ &\quad + (\text{Energy loss in liver}) \\ &= (0.6 \text{ dB/cm}) (2 \text{ cm}) + (1.2 \text{ dB/cm}) (3 \text{ cm}) \\ &\quad + (0.9 \text{ dB/cm}) (4 \text{ cm}) \\ &= 1.2 \text{ dB} + 3.6 \text{ dB} + 3.6 \text{ dB} \\ &= 8.4 \text{ dB}\end{aligned}$$

For an ultrasound beam that traverses the block of tissue and, after reflection, returns through the tissue block, the total attenuation is twice 8.4 dB or 16.8 dB.

REFLECTION

In most diagnostic applications of ultrasound, use is made of ultrasound waves reflected from interfaces between different tissues in the patient. The fraction of the impinging energy reflected from an interface depends on the difference in acoustic impedance of the media on opposite sides of the interface.

The acoustic impedance Z of a medium is the product of the density ρ of the medium and the velocity of ultrasound in the medium:

$$Z = \rho c$$

Acoustic impedances of several materials are listed in the margin. For an ultrasound wave incident perpendicularly upon an interface, the fraction α_R of the incident energy that is reflected (i.e., the reflection coefficient α_R) is

$$\alpha_R = \left(\frac{Z_2 - Z_1}{Z_2 + Z_1} \right)^2$$

where Z_1 and Z_2 are the acoustic impedances of the two media. The fraction of the incident energy that is transmitted across an interface is described by the transmission coefficient α_T , where

$$\alpha_T = \frac{4Z_1Z_2}{(Z_1 + Z_2)^2}$$

Obviously $\alpha_T + \alpha_R = 1$.

Attenuation causes a loss of signal intensity when structures at greater depths are imaged with ultrasound. This loss of intensity is compensated by increasing the amplification of the signals.

Differences in attenuation among tissues causes enhancement and shadowing in ultrasound images.

In this discussion, reflection is assumed to occur at interfaces that have dimensions greater than the ultrasound wavelength. In this case, the reflection is termed *specular reflection*.

Acoustic impedance may be expressed in units of rayls, where a rayl = $1 \text{ kg} \cdot \text{m}^{-2} \cdot \text{sec}^{-1}$.

The rayl is named for Lord Rayleigh [John Strutt (1842–1919), the 3rd Baron Rayleigh], a British physicist who pioneered the study of molecular motion in gasses that explains sound propagation.

Approximate Acoustic Impedances of Selected Materials

Material	Acoustic Impedance ($\text{kg}\cdot\text{m}^{-2}\cdot\text{sec}^{-1}$) $\times 10^{-4}$
Air at standard temperature and pressure	0.0004
Water	1.50
Polyethylene	1.85
Plexiglas	3.20
Aluminum	18.0
Mercury	19.5
Brass	38.0
Fat	1.38
Aqueous and vitreous humor of eye	1.50
Brain	1.55
Blood	1.61
Kidney	1.62
Human soft tissue, mean value	1.63
Spleen	1.64
Liver	1.65
Muscle	1.70
Lens of eye	1.85
Skull bone	6.10

With a large impedance mismatch at an interface, much of the energy of an ultrasound wave is reflected, and only a small amount is transmitted across the interface. For example, ultrasound beams are reflected strongly at air–tissue and air–water interfaces because the impedance of air is much less than that of tissue or water.

Example 19-3

At a “liver–air” interface, $Z_1 = 1.65$ and $Z_2 = 0.0004$ (both multiplied by 10^{-4} with units of $\text{kg}\cdot\text{m}^{-2}\cdot\text{sec}^{-1}$).

$$\alpha_R = \left(\frac{1.65 - 0.0004}{1.65 + 0.0004} \right)^2, \quad \alpha_T = \frac{4(1.65)(0.0004)}{(1.65 + 0.0004)^2}$$

$$= 0.9995 \quad \quad \quad = 0.0005$$

Thus 99.95% of the ultrasound energy is reflected at the air–liver interface, and only 0.05% of the energy is transmitted. At a muscle ($Z = 1.70$)–liver ($Z = 1.65$) interface,

$$\alpha_R = \left(\frac{1.70 - 1.65}{1.70 + 1.65} \right)^2, \quad \alpha_T = \frac{4(1.70)(1.65)}{(1.70 + 1.65)^2}$$

$$= 0.015 \quad \quad \quad = 0.985$$

At a muscle–liver interface, slightly more than 1% of the incident energy is reflected, and about 99% of the energy is transmitted across the interface. Even though the reflected energy is small, it is often sufficient to reveal the liver border. The magnitudes of echoes from various interfaces in the body are described in Figure 19-5.

Because of the high value of the coefficient of ultrasound reflection at an air–tissue interface, water paths and various creams and gels are used during ultrasound examinations to remove air pockets (i.e., to obtain good acoustic coupling) between the ultrasound transducer and the patient’s skin. With adequate acoustic coupling,

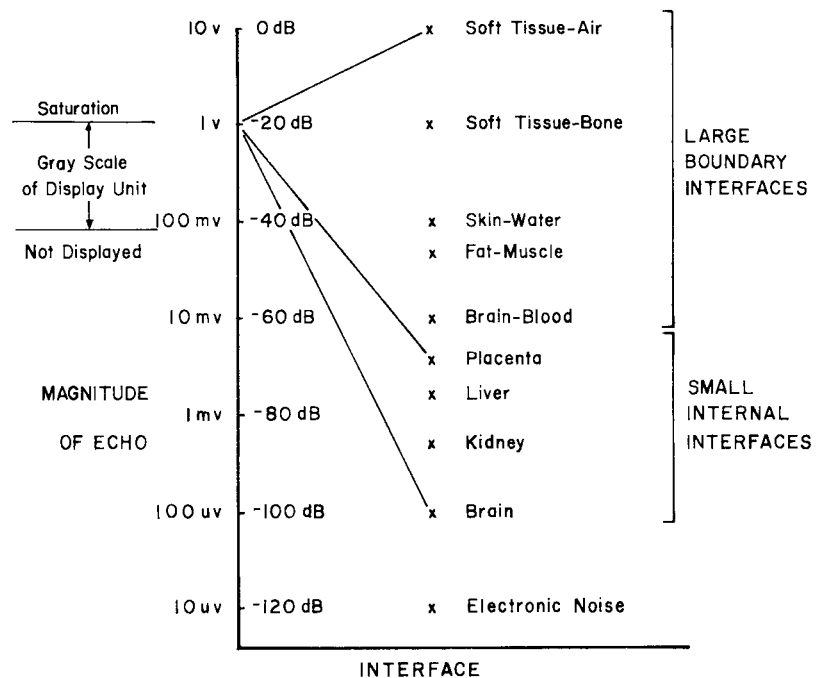


FIGURE 19-5

Range of echoes from biologic interfaces and selection of internal echoes to be displayed over the major portion of the gray scale in an ultrasound unit. (From Kossoff, G., et al.¹² Used with permission).

the ultrasound waves will enter the patient with little reflection at the skin surface. Similarly, strong reflections of ultrasound occur at the boundary between the chest wall and the lungs and at the millions of air–tissue interfaces within the lungs. Because of the large impedance mismatch at these interfaces, efforts to use ultrasound as a diagnostic tool for the lungs have been unrewarding. The impedance mismatch is also high between soft tissues and bone, and the use of ultrasound to identify tissue characteristics in regions behind bone has had limited success.

The discussion of ultrasound reflection above assumes that the ultrasound beam strikes the reflecting interface at a right angle. In the body, ultrasound impinges upon interfaces at all angles. For any angle of incidence, the angle at which the reflected ultrasound energy leaves the interface equals the angle of incidence of the ultrasound beam; that is,

$$\text{Angle of incidence} = \text{Angle of reflection}$$

In a typical medical examination that uses reflected ultrasound and a transducer that both transmits and detects ultrasound, very little reflected energy will be detected if the ultrasound strikes the interface at an angle more than about 3 degrees from perpendicular. A smooth reflecting interface must be essentially perpendicular to the ultrasound beam to permit visualization of the interface.

REFRACTION

As an ultrasound beam crosses an interface obliquely between two media, its direction is changed (i.e., the beam is bent). If the velocity of ultrasound is higher in the second medium, then the beam enters this medium at a more oblique (less steep) angle. This behavior of ultrasound transmitted obliquely across an interface is termed *refraction*. The relationship between incident and refraction angles is described by Snell's law:

$$\frac{\text{Sine of incidence angle}}{\text{Sine of refractive angle}} = \frac{\text{Velocity in incidence medium}}{\text{Velocity in refractive medium}} \quad (19-4)$$

$$\frac{\sin \theta_i}{\sin \theta_r} = \frac{c_i}{c_r}$$

For example, an ultrasound beam incident obliquely upon an interface between muscle (velocity 1580 m/sec) and fat (velocity 1475 m/sec) will enter the fat at a steeper angle.

If an ultrasound beam impinges very obliquely upon a medium in which the ultrasound velocity is higher, the beam may be refracted so that no ultrasound energy enters the medium. The incidence angle at which refraction causes no ultrasound to enter a medium is termed the *critical angle* θ_c . For the critical angle, the angle of refraction is 90 degrees, and the sine of 90 degrees is 1. From Eq. (19-4),

$$\frac{\sin \theta_c}{\sin 90^\circ} = \frac{c_i}{c_r}$$

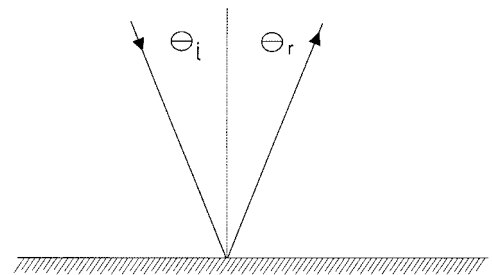
but

$$\sin 90^\circ = 1$$

therefore

$$\theta_c = \sin^{-1}[c_i/c_r]$$

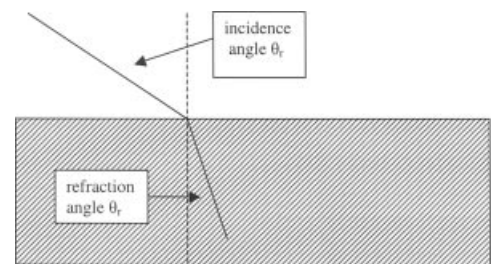
where $\sin^{-1}$, or arcsin, refers to the angle whose sine is c_i/c_r . For any particular interface, the critical angle depends only upon the velocity of ultrasound in the two media separated by the interface.



MARGIN FIGURE 19-3

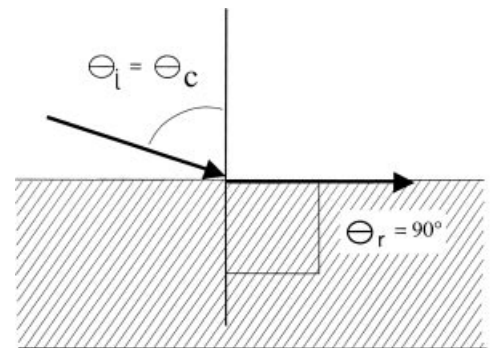
Ultrasound reflection at an interface, where the angle of incidence θ_i equals the angle of reflection θ_r .

Two conditions are required for refraction to occur: (1) The sound beam must strike an interface at an angle other than 90° ; (2) the speed of sound must differ on opposite sides of the interface.



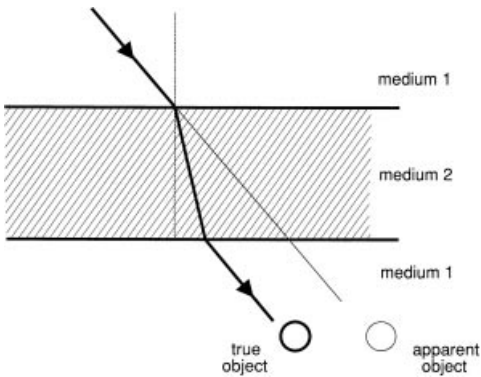
MARGIN FIGURE 19-4

Refraction of ultrasound at an interface, where the ratio of the velocities of ultrasound in the two media is related to the sine of the angles of incidence and refraction.



MARGIN FIGURE 19-5

For an incidence angle θ_c equal to the critical angle, refraction causes the sound to be transmitted along the surface of the material. For incidence angles greater than θ_c , sound transmission across the interface is prevented by refraction.



MARGIN FIGURE 19-6
Lateral displacement of an ultrasound beam as it traverses a slab interposed in an otherwise homogeneous medium.

Maximum ultrasound intensities (mW/cm²) recommended by the U.S. Food and Drug Administration for various diagnostic applications. The values are spatial-peak, temporal-average (SPTA) values.^{5,13}

Use	(Intensity) _{max}
Cardiac	430
Peripheral vessels	720
Ophthalmic	17
Abdominal	94
Fetal	94

American Institute of Ultrasound in Medicine Safety Statement on Diagnostic Ultrasound has been in use since the early 1950s. Given its known benefits and recognized efficacy for medical diagnosis, including use during human pregnancy, the American Institute of Ultrasound in Medicine hereby addresses the clinical safety of such use as follows:

No confirmed biological effects on patients or instrument operators caused by exposure at intensities typical of present diagnostic ultrasound instruments have ever been reported. Although the possibility exists that such biological effects may be identified in the future, current data indicate that the benefits to patients of the prudent use of diagnostic ultrasound outweigh the risks, if any, that may be present.

Refraction is a principal cause of artifacts in clinical ultrasound images. In Margin Figure 19-6, for example, the ultrasound beam is refracted at a steeper angle as it crosses the interface between medium 1 and 2 ($c_1 > c_2$). As the beam emerges from medium 2 and reenters medium 1, it resumes its original direction of motion. The presence of medium 2 simply displaces the ultrasound beam laterally for a distance that depends upon the difference in ultrasound velocity and density in the two media and upon the thickness of medium 2. Suppose a small structure below medium 2 is visualized by reflected ultrasound. The position of the structure would appear to the viewer as an extension of the original direction of the ultrasound through medium 1. In this manner, refraction adds spatial distortion and resolution loss to ultrasound images.

■ ABSORPTION

Relaxation processes are the primary mechanisms of energy dissipation for an ultrasound beam transversing tissue. These processes involve (a) removal of energy from the ultrasound beam and (b) eventual dissipation of this energy primarily as heat. As discussed earlier, ultrasound is propagated by displacement of molecules of a medium into regions of compression and rarefaction. This displacement requires energy that is provided to the medium by the source of ultrasound. As the molecules attain maximum displacement from an equilibrium position, their motion stops, and their energy is transformed from kinetic energy associated with motion to potential energy associated with position in the compression zone. From this position, the molecules begin to move in the opposite direction, and potential energy is gradually transformed into kinetic energy. The maximum kinetic energy (i.e., the highest molecular velocity) is achieved when the molecules pass through their original equilibrium position, where the displacement and potential energy are zero. If the kinetic energy of the molecule at this position equals the energy absorbed originally from the ultrasound beam, then no dissipation of energy has occurred, and the medium is an ideal transmitter of ultrasound. Actually, the conversion of kinetic to potential energy (and vice versa) is always accompanied by some dissipation of energy. Therefore, the energy of the ultrasound beam is gradually reduced as it passes through the medium. This reduction is termed *relaxation energy loss*. The rate at which the beam energy decreases is a reflection of the attenuation properties of the medium.

The effect of frequency on the attenuation of ultrasound in different media is described in Table 19-5.^{14–18} Data in this table are reasonably good estimates of the influence of frequency on ultrasound absorption over the range of ultrasound frequencies used diagnostically. However, complicated structures such as tissue samples often exhibit a rather complex attenuation pattern for different frequencies, which probably reflects the existence of a variety of relaxation frequencies and other molecular energy absorption processes that are poorly understood at present. These complex attenuation patterns are reflected in the data in Figure 19-6.

TABLE 19-5 Variation of Ultrasound Attenuation Coefficient α with Frequency in Megahertz, Where α_1 Is the Attenuation Coefficient at 1 MHz

Tissue	Frequency Variation	Material	Frequency Variation
Blood	$\alpha = \alpha_1 \times v$	Lung	$\alpha = \alpha_1 \times v$
Fat	$\alpha = \alpha_1 \times v$	Liver	$\alpha = \alpha_1 \times v$
Muscle (across fibers)	$\alpha = \alpha_1 \times v$	Brain	$\alpha = \alpha_1 \times v$
Muscle (along fibers)	$\alpha = \alpha_1 \times v$	Kidney	$\alpha = \alpha_1 \times v$
Aqueous and vitreous humor of eye	$\alpha = \alpha_1 \times v$	Spinal cord	$\alpha = \alpha_1^1 \times v$
Lens of eye	$\alpha = \alpha_1 \times v$	Water	$\alpha = \alpha_1 \times v^2$
Skull bone	$\alpha = \alpha_1 \times v^2$	Caster oil	$\alpha = \alpha_1 \times v^2$
		Lucite	$\alpha = \alpha_1 \times v$

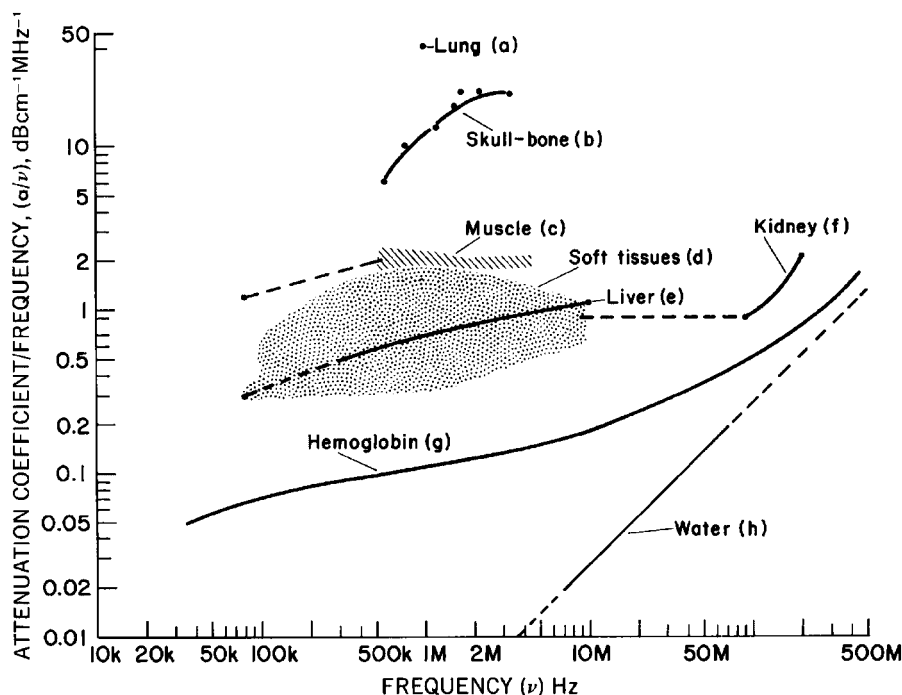


FIGURE 19-6

Ultrasound attenuation coefficient as a function of frequency for various tissue samples.

(From Wells, P. N. T.¹⁸ Used with permission.)

If gas bubbles are present in a material through which a sound wave is passing, the compressions and rarefactions cause the bubble to shrink and expand in resonance with the sound wave. The oscillation of such bubbles is referred to as *stable cavitation*. Stable cavitation is not a major mechanism for absorption at ultrasound intensities used diagnostically, but it can be a significant source of scatter.

If an ultrasound beam is intense enough and of the right frequency, the ultrasound-induced mechanical disturbance of the medium can be so great that microscopic bubbles are produced in the medium. The bubbles are formed at foci, such as molecules in the rarefaction zones, and may grow to a cubic millimeter or so in size. As the pressure in the rarefaction zone increases during the next phase of the ultrasound cycle, the bubbles shrink to 10^{-2} mm^3 or so and collapse, thereby creating minute shock waves that seriously disturb the medium if produced in large quantities. The effect, termed *dynamic cavitation*, produces high temperatures (up to $10,000^\circ\text{C}$) at the point where the collapse occurs.¹⁹ Dynamic cavitation is associated with absorption of energy from the beam. Free radicals are also produced in water surrounding the collapse. Dynamic cavitation is not a significant mechanism of attenuation at diagnostic intensities, although there is evidence that it may occur under certain conditions.²⁰

Dynamic cavitation is also termed “transient cavitation.”

SUMMARY

- Properties of ultrasound waves include:
 - Compression and rarefaction
 - Requires a transmissive medium
 - Constructive and destructive interference
- The relative intensity and pressure of ultrasound waves are described in units of decibels.

- Ultrasound may be reflected or refracted at a boundary between two media. These properties are determined by the angle of incidence of the ultrasound and the impedance mismatch at the boundary.
- Energy may be removed from an ultrasound beam by various processes, including relaxation energy loss.
- The presence of gas bubbles in a medium may give rise to stable and dynamic cavitation.

PROBLEMS

- 19-1. Explain what is meant by a longitudinal wave, and describe how an ultrasound wave is propagated through a medium.
- *19-2. An ultrasound beam is attenuated by a factor of 20 in passing through a medium. What is the attenuation of the medium in decibels?
- *19-3. Determine the fraction of ultrasound energy transmitted and reflected at interfaces between (a) fat and muscle and (b) lens and aqueous and vitreous humor of the eye.
- *19-4. What is the angle of refraction for an ultrasound beam incident at an angle of 15 degrees from muscle into bone?
- 19-5. Explain why refraction contributes to resolution loss in ultrasound imaging.
- *19-6. A region of tissue consists of 3 cm fat, 2 cm muscle (ultrasound propagated parallel to fibers), and 3 cm liver. What is the approximate total energy loss of ultrasound in the tissue?

*For those problems marked with an asterisk, answers are provided on p. 493.

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OBJECTIVES

After studying this chapter, the reader should be able to:

- Explain the piezoelectric effect and its use in ultrasound transducers.
- Characterize the properties of an ultrasound transducer, including those that influence the resonance frequency.
- Describe the properties of an ultrasound beam, including the Fresnel and Fraunhofer zones.
- Delineate the characteristics of focused ultrasound beams and various ultrasound probes.
- Identify different approaches to multitransducer arrays and the advantages of each.

INTRODUCTION

A transducer is any device that converts one form of energy into another. An ultrasound transducer converts electrical energy into ultrasound energy and vice versa. Transducers for ultrasound imaging consist of one or more piezoelectric crystals or elements. The basic properties of ultrasound transducers (resonance, frequency response, focusing, etc.) can be illustrated in terms of single-element transducers. However, imaging is often preformed with multiple-element “arrays” of piezoelectric crystals.

PIEZOELECTRIC EFFECT

The piezoelectric effect was first described by Pierre and Jacques Curie in 1880.

The movement of the surface of a piezoelectric crystal used in diagnostic imaging is on the order of a few micrometers (10^{-3} mm) at a rate of several million times per second. This movement, although not discernible to the naked eye, is sufficient to transmit ultrasound energy into the patient.

An ultrasound transducer driven by a continuous alternating voltage produces a continuous ultrasound wave. Continuous-wave (CW) transducers are used in CW Doppler ultrasound. A transducer driven by a pulsed alternating voltage produces ultrasound bursts that are referred to collectively as pulse-wave ultrasound. Pulsed ultrasound is used in most applications of ultrasound imaging. Pulsed Doppler uses ultrasound pulses of longer duration than those employed in pulse-wave imaging.

The *piezoelectric effect* is exhibited by certain crystals that, in response to applied pressure, develop a voltage across opposite surfaces.^{1–3} This effect is used to produce an electrical signal in response to incident ultrasound waves. The magnitude of the electrical signal varies directly with the wave pressure of the incident ultrasound. Similarly, application of a voltage across the crystal causes deformation of the crystal—either compression or extension depending upon the polarity of the voltage. This deforming effect, termed the *converse piezoelectric effect*, is used to produce an ultrasound beam from a transducer.

Many crystals exhibit the piezoelectric effect at low temperatures, but are unsuitable as ultrasound transducers because their piezoelectric properties do not exist at room temperature. The temperature above which a crystal's piezoelectric properties disappear is known as the *Curie point* of the crystal.

A common definition of the efficiency of a transducer is the fraction of applied energy that is converted to the desired energy mode. For an ultrasound transducer, this definition of efficiency is described as the electromechanical coupling coefficient k_c . If mechanical energy (i.e., pressure) is applied, we obtain

$$k_c^2 = \frac{\text{Mechanical energy converted to electrical energy}}{\text{Applied mechanical energy}}$$

If electrical energy is applied, we obtain

$$k_c^2 = \frac{\text{Electrical energy converted to mechanical energy}}{\text{Applied electrical energy}}$$

Values of k_c for selected piezoelectric crystals are listed in Table 20-1.

Essentially all diagnostic ultrasound units use piezoelectric crystals for the generation and detection of ultrasound. A number of piezoelectric crystals occur in nature (e.g., quartz, Rochelle salts, lithium sulfate, tourmaline, and ammonium dihydrogen phosphate [ADP]). However, crystals used clinically are almost invariably man-made

TABLE 20-1 Properties of Selected Piezoelectric Crystals

Materials	Electromechanical Coupling Coefficient (K_c)	Curie Point (°C)
Quartz	0.11	550
Rochelle salt	0.78	45
Barium titanate	0.30	120
Lead zirconate titanate (PZT-4)	0.70	328
Lead zirconate titanate (PZT-5)	0.70	365

ceramic ferroelectrics. The most common man-made crystals are barium titanate, lead metaniobate, and lead zirconate titanate (PZT).

■ TRANSDUCER DESIGN

The piezoelectric crystal is the functional component of an ultrasound transducer. A crystal exhibits its greatest response at the *resonance frequency*. The resonance frequency is determined by the thickness of the crystal (the dimension of the crystal along the axis of the ultrasound beam). As the crystal goes through a complete cycle from contraction to expansion to the next contraction, compression waves move toward the center of the crystal from opposite crystal faces. If the crystal thickness equals one wavelength of the sound waves, the compressions arrive at the opposite faces just as the next crystal contraction begins. The compression waves oppose the contraction and “dampen” the crystal’s response. Therefore it is difficult (i.e., energy would be wasted) to “drive” a crystal with a thickness of one wavelength. If the crystal thickness equals half of the wavelength, a compression wave reaches the opposite crystal face just as expansion is beginning to occur. Each compression wave produced in the contraction phase aids in the expansion phase of the cycle. A similar result is obtained for any odd multiple of half wavelengths (e.g., $3\lambda/2$, $5\lambda/2$), with the crystal progressing through more than one cycle before a given compression wave arrives at the opposite face. Additional crystal thickness produces more attenuation, so the most efficient operation is achieved for a crystal with a thickness equal to half the wavelength of the desired ultrasound. A crystal of half-wavelength thickness resonates at a frequency ν :

$$\begin{aligned}\nu &= \frac{c}{\lambda} \\ &= \frac{c}{2t}\end{aligned}$$

where $\lambda = 2t$

Example 20-1

For a 1.5-mm-thick quartz disk (velocity of ultrasound in quartz = 5740 m/sec), what is the resonance frequency?

$$\begin{aligned}\nu &= \frac{5740 \text{ m/sec}}{2(0.0015 \text{ m})} \\ &= 1.91 \text{ MHz}\end{aligned}$$

To establish electrical contact with a piezoelectric crystal, faces of the crystal are coated with a thin conducting film, and electric contacts are applied. The crystal is mounted at one end of a hollow metal or metal-lined plastic cylinder, with the front face of the crystal coated with a protective plastic that provides efficient transfer of sound between the crystal and the body. The plastic coating at the face of the crystal has

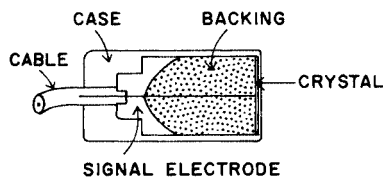
In some transducers of newer design, the piezoelectric ceramic is mixed with epoxy to form a *composite ceramic*. Composite ceramics have several performance advantages in comparison with conventional ceramics.⁴

The components of an ultrasound transducer include the

- Piezoelectric crystal
- Damping material
- Electrodes
- Housing
- Matching layer
- Insulating cover

The transducer described here is a “single-element” transducer. Such a transducer is used in some ophthalmological, m-mode, and pulsed Doppler applications. Most other applications of ultrasound employ one of the multielement transducers described later in this chapter.

High-frequency ultrasound transducers employ thin (<1 mm) piezoelectric crystals. A thicker crystal yields ultrasound of lower frequency.

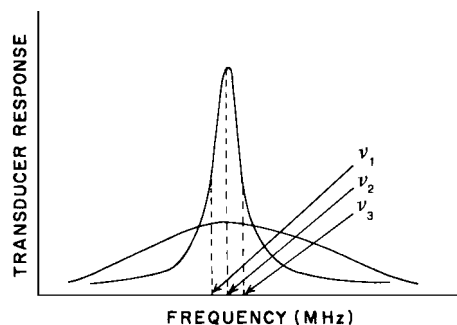
**MARGIN FIGURE 20-1**

Typical ultrasound transducer.

Damping the reverberation of an ultrasound transducer is similar to packing foam rubber around a ringing bell.

The few reverberation cycles after each voltage pulse applied to a damped ultrasound crystal is described as the *ringdown* of the crystal.

Ultrasound frequencies from 2–10 MHz are used for most diagnostic medical applications.

**MARGIN FIGURE 20-2**

Frequency–response curves for undamped (sharp curve) and damped (broad curve) transducers.

a thickness of $\frac{1}{4}\lambda$ and is called a *quarter-wavelength* matching layer. A $\frac{1}{4}\lambda$ thickness maximizes energy transfer from the transducer to the patient. An odd multiple of one-quarter wavelengths would perform the same function, but the greater thickness of material would increase attenuation. Therefore, a single one-quarter wavelength thickness commonly is used for the matching layer. The front face of the crystal is connected through the cylinder to ground potential. The remainder of the crystal is electrically and acoustically insulated from the cylinder.

With only air behind the crystal, ultrasound transmitted into the cylinder from the crystal is reflected from the cylinder's opposite end. The reflected ultrasound reinforces the ultrasound propagated in the forward direction from the transducer. This reverberation of ultrasound in the transducer itself contributes energy to the ultrasound beam. It also extends the pulse duration (the time over which the ultrasound pulse is produced). Extension of the pulse duration (sometimes called temporal pulse length) is no problem in some clinical uses of ultrasound such as continuous-wave and pulsed-Doppler applications. For these purposes, ultrasound probes with air-backed crystals may be used. However, most ultrasound imaging applications utilize short pulses of ultrasound, and suppression of ultrasound reverberation in the transducer is desirable. Suppression or “damping” of reverberation is accomplished by filling the transducer cylinder with a backing material such as tungsten powder embedded in epoxy resin. Sometimes, rubber is added to the backing to increase the absorption of ultrasound. Often, the rear surface of the backing material is sloped to prevent direct reflection of ultrasound pulses back to the crystal. The construction of a typical ultrasound transducer is illustrated in the margin. The crystal may be flat, as shown in the drawing, or curved to focus the ultrasound beam.

As a substitute for physical damping with selected materials placed behind the crystal, electronic damping may be used. In certain applications, including those that use a small receiving transducer, a resistor connecting the two faces of an air-backed crystal may provide adequate damping. Another approach, termed *dynamic damping*, uses an initial electrical pulse to stimulate the transducer, followed immediately by a voltage pulse of opposite polarity to suppress continuation of transducer action.

FREQUENCY RESPONSE OF TRANSDUCERS

An ultrasound transducer is designed to be maximally sensitive to ultrasound of a particular frequency, termed the *resonance frequency* of the transducer. The resonance frequency is determined principally by the thickness of the piezoelectric crystal. Thin crystals yield high resonance frequencies, and vice versa. The resonance frequency is revealed by a curve of transducer response plotted as a function of ultrasound frequency. In the illustration in the margin, the frequency response characteristics of two transducers are illustrated. The curve for the undamped transducer displays a sharp frequency response over a limited frequency range. Because of greater energy absorption in the damped transducer, the frequency response is much broader and not so sharply peaked at the transducer resonance frequency. On the curve for the undamped transducer, points ν_1 and ν_3 represent frequencies on either side of the resonance frequency where the response has diminished to half. These points are called the half-power points, and they encompass a range of frequencies termed the bandwidth of the transducer. The ratio of the “center,” or resonance frequency ν_2 , to the bandwidth ($\nu_3 - \nu_1$) is termed the *Q* value of the transducer. The *Q* value describes the sharpness of the frequency response curve, with a high *Q* value indicating a sharply-peaked frequency response.

$$Q \text{ value} = \frac{\nu_2}{\nu_3 - \nu_1}$$

Transducers used in ultrasound imaging must furnish short ultrasound pulses and respond to returning echoes over a wide range of frequencies. For these reasons, heavily damped transducers with low *Q* values (e.g., 2 to 3) are usually desired.

Because part of the damping is provided by the crystal itself, crystals such as PZT (lead zirconate titanate) or lead metaniobate with high internal damping and low Q values are generally preferred for imaging.

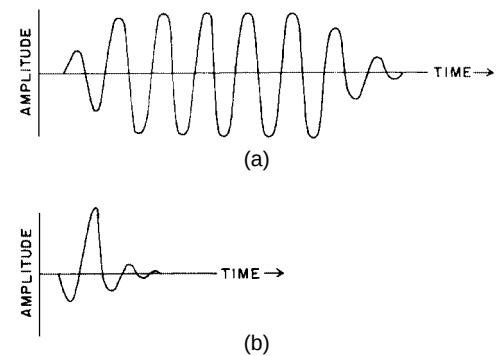
The efficiency with which an ultrasound beam is transmitted from a transducer into a medium (and vice versa) depends on how well the transducer is coupled to the medium. If the acoustic impedance of the coupling medium is not too different from that of either the transducer or the medium and if the thickness of the coupling medium is much less than the ultrasound wavelength, then the ultrasound is transmitted into the medium with little energy loss. Transmission with almost no energy loss is accomplished, for example, with a thin layer of oil placed between transducer and skin during a diagnostic ultrasound examination. Transmission with minimum energy loss occurs when the impedance of the coupling medium is intermediate between the impedances of the crystal and the medium. The ideal impedance of the coupling medium is

$$Z_{\text{coupling medium}} = \sqrt{Z_{\text{transducer}} \times Z_{\text{medium}}}$$

Two methods are commonly used to generate ultrasound beams. For continuous-wave beams, an oscillating voltage is applied with a frequency equal to that desired for the ultrasound beam. A similar voltage of prescribed duration is used to generate long pulses of ultrasound energy, as shown in the margin (A). For clinical ultrasound imaging, short pulses usually are preferred. These pulses are produced by shocking the crystal into mechanical oscillation by a momentary change in the voltage across the crystal. The oscillation is damped quickly by the methods described earlier to furnish ultrasound pulses as short as half a cycle. The duration of a pulse usually is defined as the number of half cycles in the pulse with an amplitude greater than one fourth of peak amplitude. The effectiveness of damping is described by the pulse dynamic range, defined as the ratio of the peak amplitude of the pulse divided by the amplitude of ripples following the pulse. A typical ultrasound pulse of short duration is illustrated in the margin (B).

The resonance frequency ν_2 is near the value posted on the imaging system for the transducer. For example, an individual “3.5”-MHz transducer may have an actual resonance frequency of 3.489 MHz.

Coupling of the transducer to the transmitting medium affects the size of the electrical signals generated by returning ultrasound pulses.



MARGIN FIGURE 20-3
Typical long (A) and short (B) ultrasound pulses.

ULTRASOUND BEAMS

Wave Fronts

The compression zones of an ultrasound wave are represented by lines perpendicular to the motion of the ultrasound wave in the medium. These lines are referred to as *wave fronts*. For an ultrasound source of large dimensions (i.e., a large-diameter transducer as compared with the wavelength), ultrasound wave fronts are represented as equally spaced straight lines such as those in Figure 20-1A. Wave fronts of this type are termed *planar wave fronts*, and the ultrasound wave they represent is termed a *planar* or *plane wave*. At the other extreme, an ultrasound wave originating from a source of very small dimensions (i.e., a point source) is represented by wave fronts that describe spheres of increasing diameter at increasing distance from the source. *Spherical wave fronts* from a point source are diagramed in Figure 20-1B.

Sources of exceptionally small or large dimensions are not routinely used in diagnostic ultrasound. Instead, sources with finite dimensions are used. These sources

Ultrasound from a point source creates spherical wave fronts. Ultrasound from a two-dimensional extended source creates planar wavefronts.

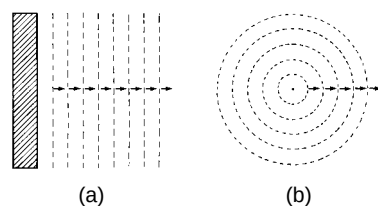
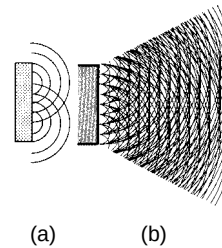


FIGURE 20-1
Ultrasound wave fronts from a source of large dimensions (A) and small dimensions (B).

**FIGURE 20-2**

A: Ultrasound sources may be considered to be a collection of point sources, each radiating spherical wavelets into the medium. **B:** Interference of the spherical wavelets establishes a characteristic pattern for the resulting net wavefronts.

The principles of interference were described by Christian Huygens (1629–1695), the Dutch mathematician, physicist, and astronomer.

Ultrasound transducer rules: (1) The length of the near field increases with increasing transducer diameter and frequency; (2) divergence in the far field decreases with increasing transducer diameter and frequency.

The length D of the Fresnel (near) zone can also be written as $D = d^2/4\lambda$, where d is the transducer diameter.

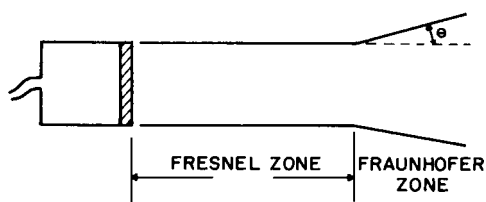
Rules for Transducer Design

For a given transducer diameter,

- the near-field length increases with increasing frequency;
- beam divergence in the far field decreases with increasing frequency

For a given transducer frequency

- the near-field length increases with increasing transducer diameter;
- beam divergence in the far field decreases with increasing transducer diameter.

**MARGIN FIGURE 20-4**

Divergence of the ultrasound beam in the Fraunhofer region. Angle θ is the Fraunhofer divergence angle.

can be considered to be a collection of point sources, each radiating spherical wave fronts (termed wavelets) into the medium, as shown in Figure 20-2A. In regions where compression zones for one wavelet intersect those of another, a condition of constructive interference is established. With constructive interference, the wavelets reinforce each other, and the total pressure in the region is the sum of the pressures contributed by each wavelet. In regions where compression zones for one wavelet intersect zones of rarefaction for another wavelet, a condition of destructive interference is established. In these regions, the molecular density is reduced.

With many spherical wavelets radiating from a transducer of reasonable size (i.e., the diameter of the transducer is considerably larger than the ultrasound wavelength), many regions of constructive and destructive interference are established in the medium. In Figure 20-2B, these regions are represented as intersections of lines depicting compression zones of individual wavelets. In this figure, the reinforcement and cancellation of individual wavelets are most noticeable in the region near the source of ultrasound. They are progressively less dramatic with increasing distance from the ultrasound source. The region near the source where the interference of wavelets is most apparent is termed the *Fresnel* (or *near*) zone. For a disk-shaped transducer of radius r , the length D of the Fresnel zone is

$$D_{\text{fresnel}} = \frac{r^2}{\lambda}$$

where λ is the ultrasound wavelength. Within the Fresnel zone, most of the ultrasound energy is confined to a beam width no greater than the transducer diameter. Beyond the Fresnel zone, some of the energy escapes along the periphery of the beam to produce a gradual divergence of the ultrasound beam that is described by

$$\sin \theta = 0.6 \left(\frac{\lambda}{r} \right)$$

where θ is the Fraunhofer divergence angle in degrees (see margin). The region beyond the Fresnel zone is termed the *Fraunhofer* (or *far*) zone.

Example 20-2

What is the length of the Fresnel zone for a 10-mm-diameter, 2-MHz unfocused ultrasound transducer?

A 10-mm-diameter transducer has a radius of 5 mm. A 2-MHz transducer has a λ of

$$\begin{aligned} \lambda &= \frac{1540 \text{ m/sec}}{2 \times 10^6/\text{sec}} = 770 \times 10^{-6} \text{ m} \\ &= 0.77 \text{ mm} \end{aligned}$$

$$\begin{aligned} D_{\text{Fresnel}} &= \frac{(5 \text{ mm})^2}{0.77 \text{ mm}} \\ &= 32.5 \text{ mm} \end{aligned}$$

TABLE 20-2 Transducer Radius and Ultrasound Frequency and Their Relationship to Fresnel Zone Depth and Beam Divergence

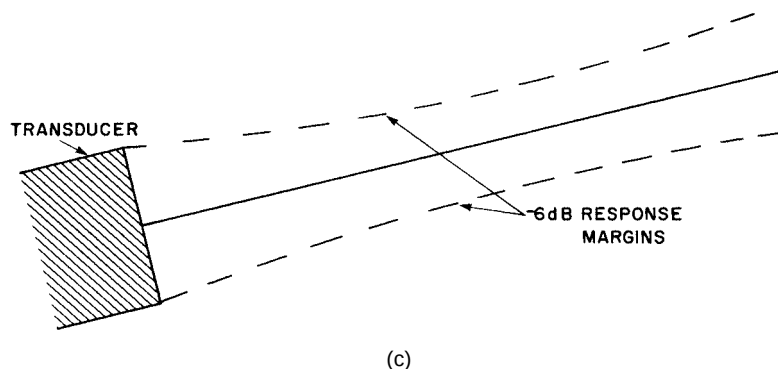
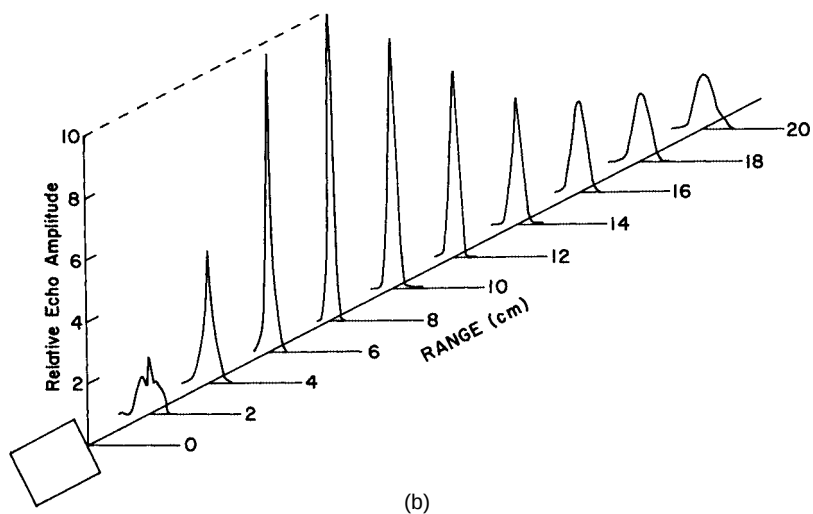
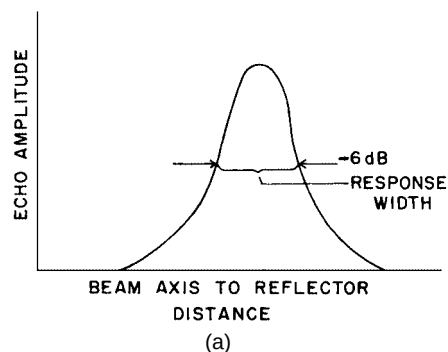
<i>Frequency (MHz)</i>	<i>Wavelength (cm)</i>	<i>Fresnel Zone Depth (cm)</i>	<i>Fraunhofer Divergence Angle (degrees)</i>
Transducer radius constant at 0.5 cm			
0.5	0.30	0.82	21.5
1.0	0.15	1.63	10.5
2.0	0.075	3.25	5.2
4.0	0.0325	6.50	2.3
8.0	0.0163	13.0	1.1
<i>Fresnel Zone Depth (cm)</i>			<i>Fraunhofer Divergence Angle in Water (degrees)</i>
<i>Radius (cm)</i>			
Frequency constant at 2 MHz			
0.25	0.83		10.6
0.5	3.33		5.3
1.0	13.33		2.6
2.0	53.33		1.3

For medical applications of ultrasound, beams with little lateral dispersion of energy (i.e., long Fresnel zones) are preferred. Hence a reasonably high ratio of transducer radius to wavelength (r/λ) is desired. This requirement can be satisfied by using ultrasound of short wavelengths (i.e., high frequencies). However, the absorption of ultrasound energy increases at higher frequencies, and frequencies for clinical imaging are limited to 2 to 20 MHz. At these frequencies a transducer radius of 10 mm or more usually furnishes an ultrasound beam with adequate directionality for medical use. The relationship of transducer radius and ultrasound frequency to the depth of the Fresnel zone and the amount of beam divergence is illustrated in Table 20-2.

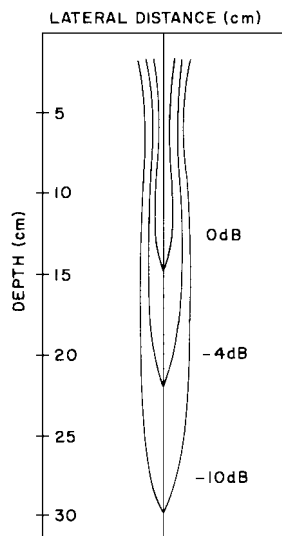
Beam Profiles

The transmission and reception patterns of an ultrasound transducer are affected by slight variations in the construction and manner of electrical stimulation of the transducer. Hence, the exact shape of an ultrasound beam is difficult to predict, and beam shapes or profiles must be measured for a particular transducer. One approach to the display of ultrasound beam characteristics is a set of pulse-echo response profiles. A profile is obtained by placing an ultrasound reflector some distance from the transducer and scanning the transducer in a direction perpendicular to the axis of the ultrasound beam. During the scan, the amplitude of the signal induced in the transducer by the reflected ultrasound is plotted as a function of the distance between the central axis of the ultrasound beam and the reflector. A pulse-echo response profile is shown in Figure 20-3A, and a set of profiles obtained at different distances from the transducer is shown in Figure 20-3B.

In Figure 20-3A, locations are indicated where the transducer response (amplitude of the reflected signal) decreases to half (-6 dB) of the response when the transducer is aligned with the reflector. The distance between these locations is termed the response width of the transducer at the particular distance (range) from the transducer. If response widths are connected between profiles at different ranges (Figure 20-3C), 6-dB response margins are obtained on each side of the ultrasound beam axis. Similarly, 20-dB response margins may be obtained by connecting the $1/10$ amplitude (-20 dB) response widths on each side of the beam axis.

**FIGURE 20-3**

A: Pulse-echo response profile and the response width of a transducer. **B:** Set of response profiles along the axis of an ultrasound beam. **C:** Response margins of -6 dB along an ultrasound beam.

**MARGIN FIGURE 20-5**

Isoecho contours for a nonfocused transducer.

Response profiles for a particular transducer are influenced by several factors, including the nature of the stimulating voltage applied to the transducer, the characteristics of the electronic circuitry of the receiver, and the shape, size, and character of the reflector. Usually, the reflector is a steel sphere or rod with a diameter of three to ten times the ultrasound wavelength. Response profiles may be distorted if the receiver electronics do not faithfully represent low-intensity signals. Some older ultrasound units cannot accurately display echo amplitudes much less than $1/10$ (-20 dB) of the largest echo recorded. These units are said to have limited dynamic range.

Another approach to describing the character of an ultrasound beam is with isoecho contours. Each contour depicts the locations of equal echo intensity for

the ultrasound beam. At each of these locations, a reflecting object will be detected with equal sensitivity. The approach usually used to measure isoecho contours is to place a small steel ball at a variety of positions in the ultrasound beam and to identify locations where the reflected echoes are equal. Connecting these locations with lines yields isoecho contours such as those in the margin, where the region of maximum sensitivity at a particular depth is labeled 0 dB, and isoecho contours of lesser intensity are labeled -4 dB, -10 dB, and so on. Isoecho contours help depict the lateral resolution of a transducer, as well as variations in lateral resolution with depth and with changes in instrument settings such as beam intensity, detector amplifier gain, and echo threshold.

Accompanying a primary ultrasound beam are small beams of greatly reduced intensity that are emitted at angles to the primary beam. These small beams, termed side lobes (see margin), are caused by vibratory modes of the transducer in the transverse plane. Side lobes can produce image artifacts in regions near the transducer, if a particularly echogenic material, such as a biopsy needle, is present.

The preceding discussion covers general-purpose, flat-surfaced transducers. For most ultrasound applications, transducers with special shapes are preferred. Among these special-purpose transducers are focused transducers, double-crystal transducers, ophthalmic probes, intravascular probes, esophageal probes, composite probes, variable-angle probes, and transducer arrays.

Focused Transducers

A focused ultrasound transducer produces a beam that is narrower at some distance from the transducer face than its dimension at the face of the transducer.^{5,6} In the region where the beam narrows (termed the focal zone of the transducer), the ultrasound intensity may be heightened by 100 times or more compared with the intensity outside of the focal zone. Because of this increased intensity, a much larger signal will be induced in a transducer from a reflector positioned in the focal zone. The distance between the location for maximum echo in the focal zone and the element responsible for focusing the ultrasound beam is termed the focal length of the transducer.

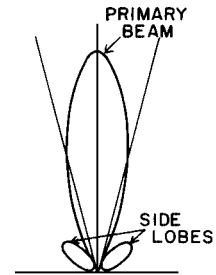
Often, the focusing element is the piezoelectric crystal itself, which is shaped like a concave disk (see figure in margin). An ultrasound beam also may be focused with mirrors and refracting lenses. Focusing lenses and mirrors are capable of increasing the intensity of an ultrasound beam by factors greater than 100. Focusing mirrors, usually constructed of tungsten-impregnated epoxy resin, are illustrated in the margin. Because the velocity of ultrasound generally is greater in a lens than in the surrounding medium, concave ultrasound lenses are focusing, and convex ultrasound lenses are defocusing (see margin). These effects are the opposite of those for the action of optical lenses on visible light. Ultrasound lenses usually are constructed of epoxy resins and plastics such as polystyrene.

For an ultrasound beam with a circular cross section, focusing characteristics such as pulse-echo response width and relative sensitivity along the beam axis depend on the wavelength of the ultrasound and on the focal length f and radius r of the transducer or other focusing element. These variables may be used to distinguish the degree of focusing of transducers by dividing the near field length r^2/λ by the focal length f . For cupped transducer faces on all but weakly focused transducers, the focal length of the transducer is equal to or slightly shorter than the radius of curvature of the transducer face. If a planoconcave lens with a radius of curvature r is attached to the transducer face, then the focal length f is

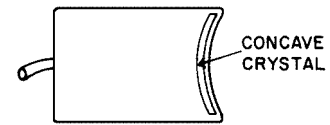
$$f = \frac{r}{1 - c_M/c_L}$$

where c_M and c_L are the velocities of ultrasound in the medium and lens, respectively.

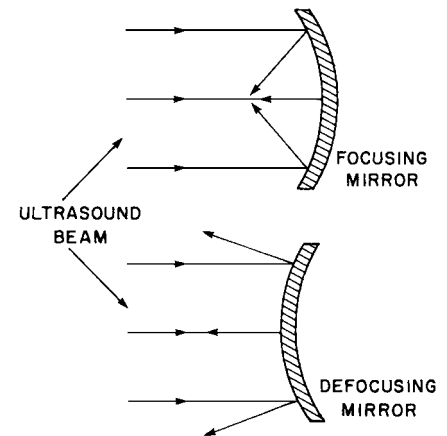
Side lobes can be reduced further by the process of *apodization*, in which the voltage applied to the transducer is diminished from the center to the periphery.



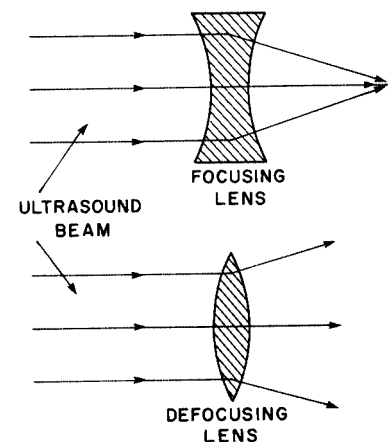
MARGIN FIGURE 20-6
Side lobes of an ultrasound beam.



MARGIN FIGURE 20-7
Focused transducer.



MARGIN FIGURE 20-8
Focusing and defocusing mirrors.

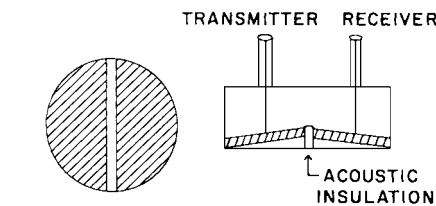


MARGIN FIGURE 20-9
Focusing and defocusing lenses.

The ratio f/d , where $d = 2r =$ the diameter of the transducer, is often described as the f -number of the transducer or other focusing element.

Degree of Focus of Transducers Expressed as a Ratio of the Near-field Length r^2/λ to the Focal Length f

Degree of Focus	Near-Field Length
	Focal Length
Weak	$(r^2/\lambda)/f \leq 1.4$
Medium weak	$1.4 < (r^2/\lambda)/f \leq 6$
Medium	$6 < (r^2/\lambda)/f \leq 20$
Strong	$20 < (r^2/\lambda)/f$



MARGIN FIGURE 20-10 Front (left) and side (right) views of a typical Doppler transducer.

The length of the focal zone of a particular ultrasound beam is the distance over which a reasonable focus and pulse-echo response are obtained. One estimate of focal zone length is

$$\text{Focal zone length} = 10\lambda \left(\frac{f}{d}\right)^2$$

where $d = 2r$ is the diameter of the transducer. These strongly focused transducers are also used for surgical applications of ultrasound where high ultrasound intensities in localized regions are needed for tissue destruction.

Doppler Probes

Transducers for continuous-wave Doppler ultrasound consist of separate transmitting and receiving crystals, usually oriented at slight angles to each other so that the transmitting and receiving areas intersect at some distance in front of the transducer (see margin). Because a sharp frequency response is desired for a Doppler transducer, only a small amount of damping material is used.

Multiple-Element Transducers

Scanning of the patient may be accomplished by physical motion of a single-element transducer. The motion of the transducer may be executed manually by the sonographer or automatically with a mechanical system. Several methods for mechanical scanning are shown in Figure 20-4. The scanning technique that uses an automatic scanning mechanism for the transducer is referred to as *mechanical sector scanning*.

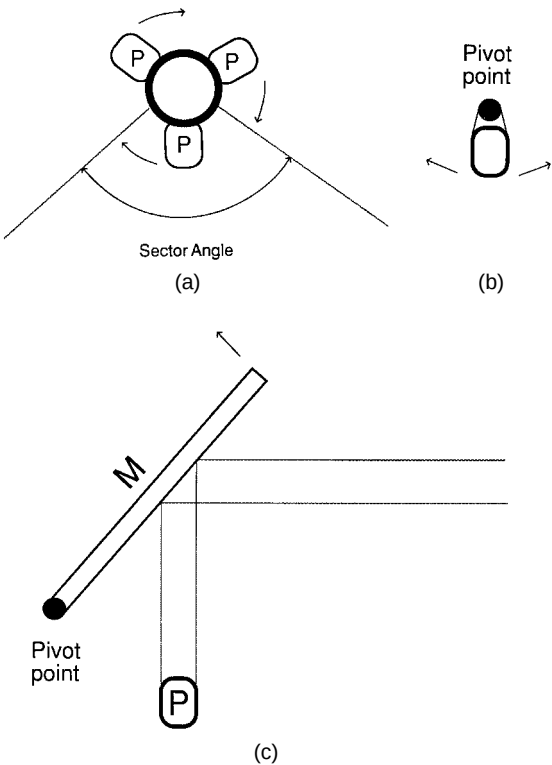


FIGURE 20-4 Three methods of mechanical scanning of the ultrasound beam. **A:** Multiple piezoelectric elements (P) mounted on a rotating head. One element is energized at a time as it rotates through the sector angle. **B:** Single piezoelectric element oscillating at high frequency. **C:** Single piezoelectric element reflected from an oscillating acoustic mirror (M).

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MARGIN FIGURE 20-11 Linear array. (From Zagzebski, J. A.⁴ Used with permission.)

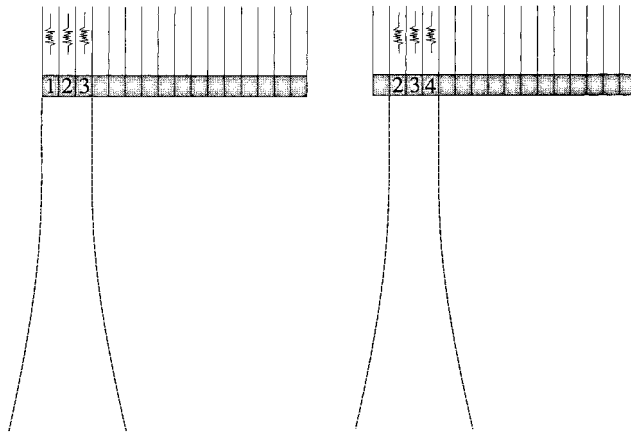


FIGURE 20-5
Electronic scanning with a linear switched array.

Alternatively, the ultrasound beam may be swept back and forth without the need for any mechanical motion through the use of transducer arrays. Transducer arrays are composed of multiple crystals that can change the direction or degree of focus of the ultrasound beam by timing the excitation of the crystals.

A linear (or curved) array of crystal elements is shown in Figure 20-5. The crystals, as many as 60 to 240 or more, are excited by voltage pulses in groups of three up to 20 or more. Each excitation of the crystal group results in a “scan line” (see next chapter). To obtain the succeeding scan line, the next crystal group is defined to overlap the first (e.g., scan line 1 is produced by crystals 1, 2, and 3, while scan line 2 is produced by crystals 2, 3, and 4, and so on). This pulsing scheme is referred to as a *linear switched array*. By sequentially exciting the entire array, an image composed of a number of scan lines (typically 64, 128, 256, etc.) is obtained.

Another method for scanning with a linear array uses *phased-array* technology. A phase array uses all (typically 128) of the elements of the array to obtain each scan line. By using slight delays between excitations of the elements, the beam may be “swept” to the left or right (Figure 20-6A). A variation in the time delay scheme in a phased array may also be used to focus the beam at various distances (Fig 20-6B) throughout each image. In this technique, called *dynamic focusing*, part of the image is acquired with the focal zone near the transducer face, and part is acquired with the focal zone farther from the transducer face. Thus, two or more images taken at different focal zones may be combined to produce a single image with better axial resolution over a broader range than is possible with a single-element transducer. *Dynamic focusing* is performed without moving parts by simply varying the timing of the excitation pulses.

The main distinction between the linear phased array and the linear switched array is that all of the elements of the phased array are used to produce each scan line while only a few of the elements of the switched array are used to produce each scan line. Linear phased arrays provide variable focus in only one dimension—that is, in the plane of the scan. Beam focus in the other dimension, the direction of slice thickness, is provided by acoustic lenses or by concavity of the elements (Figure 20-7).

Another type of array, the annular array, is capable of focusing in all planes perpendicular to the axis of the beam. The annular array (Figure 20-7) consists of a series of piezoelectric elements in the shape of concentric rings or annuli. The beam may be focused at various distances from the transducer face by varying the time delays among excitations of the rings.

Transducer Damage

Ultrasound transducers can be damaged in many ways.^{4,7} The crystals are brittle, and the wire contacts on the crystals are fragile; hence transducers should be handled

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MARGIN FIGURE 20-12

Curved (curvilinear) array. (From Zagzebski, J. A.⁴ Used with permission.)

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MARGIN FIGURE 20-13

Phased array. (From Zagzebski, J. A.⁴ Used with permission.)

Advantages of *transducer arrays*:

- They provide electronic beam steering.
- They permit dynamic focusing and beam shaping.

Transducer arrays employ the technique of a *dynamic aperture*, in which the diameter of the sensitive region of the array is expanded as signals are received from greater depths. This technique maintains a constant lateral resolution with depth.

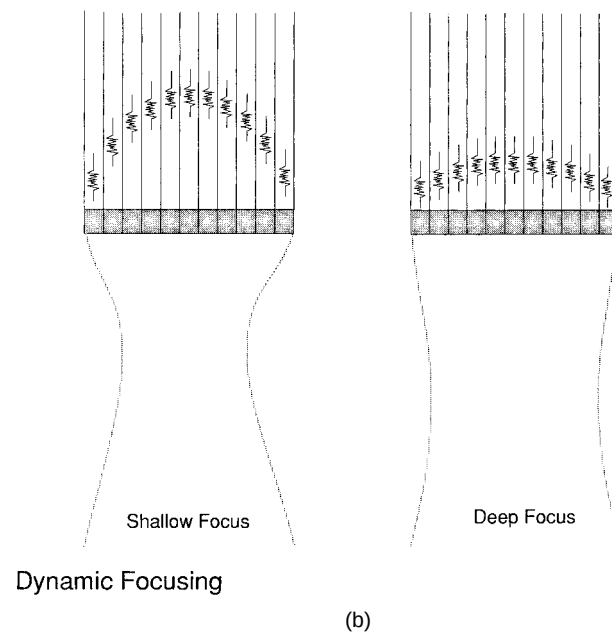
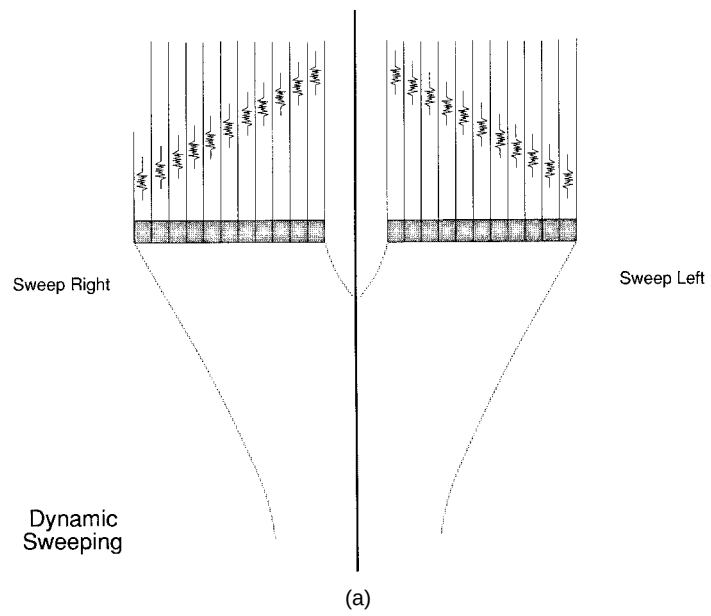


FIGURE 20-6
Electronic scanning with a linear phased array.

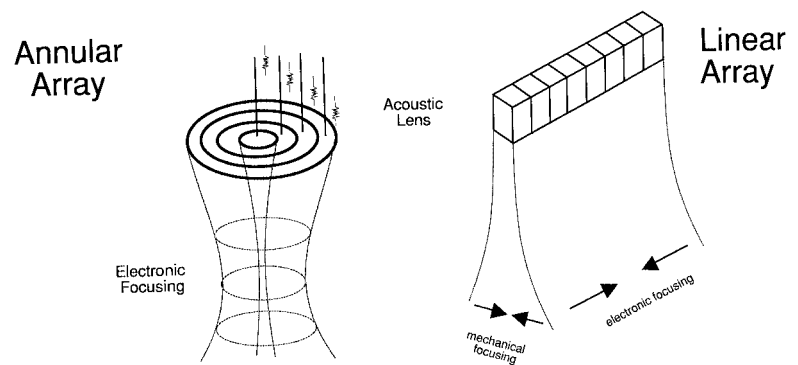


FIGURE 20-7
A linear array requires focusing in the “out-of-plane” dimension by mechanical means (lens or shape of the transducer), whereas an annular array focuses in all directions perpendicular to the axis of the beam.

carefully. Excessive voltages to the crystal should be avoided, and only watertight probes should be immersed in water. A transducer should never be heated to a temperature near the Curie point of the piezoelectric crystal. Dropping a transducer, twisting, bending, and crushing the transducer cables, and other examples of mishandling ultrasound transducers are frequent causes of dysfunction.

PROBLEMS

- 20-1. Explain what is meant by the piezoelectric effect and the converse piezoelectric effect.
- *20-2. For a piezoelectric material with an ultrasound velocity of 6000 m/sec, what thickness should a disk-shaped crystal have to provide an ultrasound beam with a frequency of 2.5 MHz?
- 20-3. What is meant by damping an ultrasound transducer, and why is this necessary? What influence does damping have on the frequency response of the transducer?
- *20-4. What is the estimated focal zone length for a 2-MHz ($\lambda = 0.075$ cm) focused ultrasound transducer with an f -number of 8?
- 20-5. A linear array can be electronically steering and focused. An annular array can be electronically focused. Can an annular array be electronically steered?

*For those problems marked with an asterisk, answers are provided on p. 493.

SUMMARY

- The active element of an ultrasound transducer is a crystal that exhibits the piezoelectric (receiver) and converse piezoelectric (transmitter) effect.
- The resonance frequency of a transducer is determined by the type of piezoelectric crystal and its thickness.
- The bandwidth of a transducer is expressed by its Q value.
- The Fresnel zone, the area of minimum beam divergence in front of the transducer, has an axial dimension determined by r^2/λ .
- The angle θ of divergence of the ultrasound beam in the Fraunhofer zone is determined by $\sin \theta = 0.6(\lambda/r)$.
- The shape of an ultrasound beam may be described by response profiles or isoecho contours.
- Real-time ultrasound scans may be obtained with mechanical sector scanners or multitransducer arrays.
- Multitransducer arrays include linear switched arrays, linear phased arrays, and annular arrays.

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